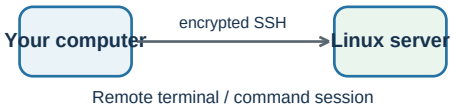
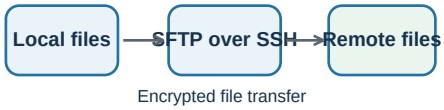
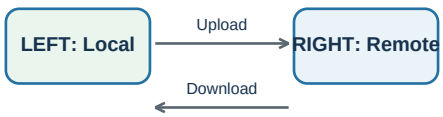
# SSH, FILEZILLA & SFTP
# Expanded Beginner Study Notes

Concepts • diagrams • commands • connection setup • practical workflows • troubleshooting • security



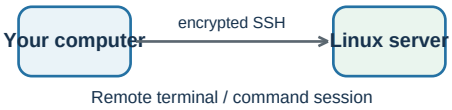




**What this guide covers:** How remote access works, how SSH authenticates users, how SFTP transfers files, how FileZilla is configured, how to use SCP and terminal SFTP, and how to diagnose common connection problems.

Written for a beginner learning Linux with Windows 11 and Ubuntu/WSL.

# 1. The big picture: local computer and remote server

A **local computer** is the device you are currently using. A **remote server** is another computer that you access over a network. It may be in a data centre, cloud platform, college lab or another location. SSH and SFTP are ways to communicate with that remote machine securely.




## Example situation

You have a project on your Windows laptop. You also have access to a Linux server. You want to log in, run Linux commands and copy your project to the server. SSH gives you a remote terminal; SFTP/FileZilla gives you a convenient way to transfer files.

## Important vocabulary

| Term | Meaning |
|---|---|
| Client | The computer or application that initiates a connection. |
| Server | The computer/service that accepts a connection. |
| Host | The server's address or name. |
| IP address | A numerical network address, such as 192.0.2.10 (example only). |
| Username | The account name used on the remote computer. |
| Port | A numbered endpoint through which a network service accepts connections. |
| Authentication | The process of proving which user or client is connecting. |
| Encryption | Scrambling network data so unauthorised parties cannot easily read it. |

# 2. What is SSH?

**SSH** means **Secure Shell**. It is a protocol for secure remote login and command execution. When you use SSH, your keyboard input and the remote terminal output travel through an encrypted connection.

## Typical use cases

• Log in to a remote Linux machine.
• Run commands without being physically at the server.
• Administer cloud virtual machines.
• Deploy and maintain applications.
• Use secure tools such as SFTP, SCP and SSH port forwarding.

## Basic command

```
ssh username@server_address
```

## Example (placeholder values)

```
ssh ubuntu@192.0.2.10
```

The command means: use SSH to connect as user `ubuntu` to the host at `192.0.2.10`. This example address is reserved for documentation and is not a server you can log into.

## What happens when you connect?

1. The client contacts the server's SSH service, commonly on TCP port 22.
2. The server presents its host key so the client can identify the server.

3. The client and server negotiate encryption and establish a secure channel.
4. The user authenticates using a password, key or another permitted method.
5. If authentication succeeds, the server opens a remote session.

# 3. SSH ports and connection syntax

| Command | Explanation |
|---|---|
| ssh user@host | Connect to the SSH service on its default port. |
| ssh -p 2222 user@host | Connect when SSH uses port 2222 instead of the default. |
| ssh -v user@host | Show verbose connection details for troubleshooting. |
| ssh -i ~/.ssh/my_key user@host | Choose a specific private-key file. |
| exit | Close the remote shell session. |

SSH normally uses TCP port **22**. A server administrator can configure another port. If the server uses a non-default port, specify it with `-p`.

# 4. Password authentication

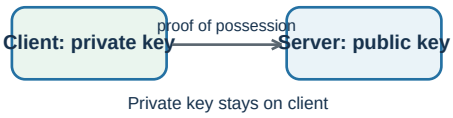
With password authentication, the server asks for the password of the remote account. When typing a password in a terminal, the screen often shows no letters, dots or asterisks. This is normal: the terminal is hiding the input.

```
ssh ubuntu@server_address
# Enter the remote account password when prompted
```

A password is not the same as your Windows login password or your WSL password unless you deliberately configured them to be the same. Use the credentials supplied for that specific server.

# 5. SSH key authentication

SSH key authentication uses a pair of mathematically related keys: a private key and a public key. The private key stays with the client. The public key is placed in the remote account's `~/.ssh/authorized_keys` file.




| Key | Where it belongs | Rule |
|---|---|---|
| Private key | Your client computer | Keep secret; protect with file permissions and optionally a passphrase. |
| Public key | Remote account's authorized_keys | Can be copied to the server; it is not secret. |

## Generate an Ed25519 key pair

```
ssh-keygen -t ed25519 -C "your_email@example.com"
```

The command asks where to save the key and whether to protect it with a passphrase. The default location is commonly `~/.ssh/id_ed25519`.

## Copy the public key to a server

```
ssh-copy-id username@server_address
```

This helper is available on many Linux systems. It typically logs in to the server and adds your public key to the remote account's authorised keys. The server must permit the required initial login and setup.

## Connect using a specific key

```
ssh -i ~/.ssh/id_ed25519 username@server_address
```

## Common SSH directory and files

| Path | Purpose |
| --- | --- |
| ~/.ssh/ | Per-user SSH configuration, keys and known-host records. |
| ~/.ssh/id_ed25519 | Example private key. |
| ~/.ssh/id_ed25519.pub | Matching public key. |
| ~/.ssh/known_hosts | Host keys for servers previously contacted. |
| ~/.ssh/authorized_keys | Public keys allowed to log in to this account on the server. |

# 6. Host keys and first-time connection warnings

The server's **host key** helps your SSH client recognise the server. On a first connection, SSH may ask whether you trust the displayed fingerprint. Verify it using a trusted source, such as your administrator or cloud console, before accepting it.

If a host key changes unexpectedly, SSH may warn that the remote host identification has changed. Do not blindly delete the warning or bypass the check. The server may have been rebuilt legitimately, but the warning can also indicate a security risk. Confirm the change first.

# 7. What are FTP, FTPS and SFTP?

| Protocol | Full form / idea | Encryption and common port |
|---|---|---|
| FTP | File Transfer Protocol | Not encrypted by default; control connection commonly TCP 21. |
| FTPS | FTP protected with TLS | Encrypted with TLS; explicit FTPS commonly starts on TCP 21. |
| SFTP | SSH File Transfer Protocol | Uses SSH encryption; commonly TCP 22. |

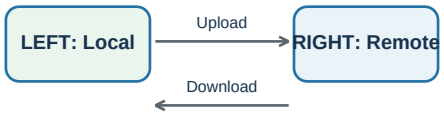
SFTP is a separate protocol that runs over SSH. It is not FTP with a security switch. SFTP can list directories, create folders, rename files, upload and download files.

## Why prefer SFTP over traditional FTP?

Traditional FTP can expose usernames, passwords and file contents to network observers. SFTP encrypts the connection through SSH, protecting authentication and transferred data in transit. Use the protocol your server supports; do not assume FTP and SFTP use the same settings.

# 8. What is FileZilla?

**FileZilla Client** is a graphical file-transfer application. It supports FTP, FTPS and SFTP. For a Linux server with SSH enabled, you will often select SFTP.




| Interface part | Meaning |
|---|---|
| Local site (left) | Your own computer's files and folders. |
| Remote site (right) | Files and folders on the connected server. |
| Filename list | Shows files and folders in the selected directory. |
| Transfer queue | Shows pending, successful and failed transfers. |
| Status/message area | Displays connection and transfer messages. |

Upload means local → remote. Download means remote → local. Dragging a file between the panels is a common way to start a transfer.

# 9. Connect FileZilla to a Linux server using SFTP

Before opening FileZilla, obtain the correct connection information from your server administrator, cloud provider or lab instructions.

| Setting | Example / explanation |
|---|---|
| Protocol | SFTP - SSH File Transfer Protocol |
| Host | Server IP address or DNS name |
| Port | 22 by default, unless the administrator specifies another port |
| Username | The account name on the remote Linux server |
| Logon type | Ask for password, key file or another supported method |
| Password / key | Credential for the remote account; do not share it. |

## Site Manager steps

1. Open FileZilla Client.
2. Select **File** → **Site Manager**.
3. Choose **New Site** and give the connection a useful name.
4. Set Protocol to **SFTP - SSH File Transfer Protocol**.
5. Enter the host, port and username.
6. Choose an authentication method supported by the server.
7. Click **Connect** and verify the host-key fingerprint before trusting it.

## Password vs key in FileZilla

With password authentication, FileZilla prompts for the remote account password. With key authentication, configure FileZilla to use the appropriate private key, following the version's supported key format and prompts. Keep the private key protected and never send it to someone else.

# 10. Upload and download safely

Example destination paths (illustrative):

```
Local Windows folder:
C:\Users\yourname\Desktop\MyProject

Remote Linux folder:
/home/ubuntu/MyProject
```

## Upload a folder

1. Navigate to the local folder in the left panel.
2. Navigate to the desired remote parent folder in the right panel.
3. Select the local folder or files.
4. Drag left-to-right or right-click and select Upload.
5. Watch the transfer queue and confirm the files appear remotely.

## Download files

1. Navigate to the remote folder in the right panel.
2. Choose a destination folder on the local side.
3. Select remote files and drag right-to-left or choose Download.
4. Check the transfer queue and confirm the local files arrived.

## Before overwriting or deleting

Check the destination path and filenames. If FileZilla asks whether to overwrite an existing file, compare timestamps and sizes before choosing. Avoid deleting remote files unless you know what they are and have permission to remove them.

# 11. Command-line file transfer with SCP

**SCP** means Secure Copy. It copies files between computers using SSH. It is useful when you prefer a terminal instead of a graphical tool.

## Upload one file from local to remote

```
scp notes.txt ubuntu@server_address:/home/ubuntu/
```

## Download one file from remote to local

```
scp ubuntu@server_address:/home/ubuntu/notes.txt .
```

The final dot means the current local directory. Use a full path if you want to save somewhere else.

## Copy a directory recursively

```
scp -r MyProject ubuntu@server_address:/home/ubuntu/
```

## Use a non-default SSH port with SCP

```
scp -P 2222 notes.txt ubuntu@server_address:/home/ubuntu/
```

> Note: SCP uses uppercase `-P` for the port option, unlike SSH's lowercase `-p`.

# 12. Command-line SFTP

## Start an SFTP session

```
sftp ubuntu@server_address
```

| Command inside SFTP | Meaning |
|---|---|
| pwd | Show the current remote directory. |
| lpwd | Show the current local directory. |
| ls | List remote files. |
| lls | List local files. |
| cd folder | Change remote directory. |
| lcd folder | Change local directory. |
| put file.txt | Upload a local file. |
| get file.txt | Download a remote file. |
| mkdir folder | Create a remote directory. |
| rename old new | Rename a remote file. |
| rm file.txt | Delete a remote file — use carefully. |
| help | Show available SFTP commands. |
| exit / bye | End the SFTP session. |

## Example session

```
sftp ubuntu@server_address
pwd
ls
put report.txt
get results.csv
exit
```

For large folder synchronisation or repeated deployments, tools such as rsync over SSH can be useful. You can learn rsync later; it is not required for this beginner topic.

# 13. SSH configuration shortcuts

If you connect to the same server frequently, you can create a client-side SSH configuration file at `~/.ssh/config`. This lets you use a short alias instead of typing all connection details every time.

```
Host my-linux-server
    HostName 192.0.2.10
    User ubuntu
    Port 22
    IdentityFile ~/.ssh/id_ed25519
```

Then connect with:

```
ssh my-linux-server
```

The example IP is illustrative. Keep the config file readable only by your user where appropriate; a common permission is `chmod 600 ~/.ssh/config`.

# 14. Permissions that affect SFTP

SFTP uses the permissions of the authenticated remote Linux account. Being able to log in does not mean you can write to every directory.

| Situation | What it means |
|---|---|
| Can list a directory | The account has sufficient directory read and execute permissions. |
| Can upload into a directory | The account needs write and execute permission on the destination directory. |
| Can download a file | The account needs permission to read the file and traverse its parent directories. |
| Permission denied | The account may lack the required permission, or a server policy may restrict the operation. |

Do not use `chmod 777` as a quick fix. Ask the administrator or inspect ownership and permissions; change only the specific permissions required.

# 15. Troubleshooting common errors

| Error / symptom | Possible cause and first check |
|---|---|
| Connection timed out | Wrong host/port, network route or firewall; confirm server address and that port is reachable. |
| Connection refused | Host is reachable but no service is accepting connections on that port, or a rule is rejecting it. |
| Permission denied (publickey) | Wrong key, public key not installed for that account, wrong username or key authentication not allowed. |
| Permission denied (password) | Wrong password, wrong username or password login disabled. |
| Host key verification failed | Known host key differs or cannot be verified; confirm the server identity before changing known_hosts. |
| SFTP login works but upload fails | Destination directory may not be writable by the remote account. |
| FileZilla cannot connect | Check protocol is SFTP, host, port, username, authentication and server SSH availability. |

## Useful diagnostic command

```
ssh -v username@server_address
```

Verbose mode shows more connection details. Read the messages carefully and do not paste passwords, private keys or other secrets into support chats.

# 16. Security best practices

• Use SFTP or another encrypted transfer method instead of traditional unencrypted FTP over untrusted networks.
• Keep private keys secret and protect them with appropriate file permissions and a passphrase where practical.
• Verify SSH host-key fingerprints, especially on first connection or after a host-key warning.
• Use a separate account with only the permissions needed for the task.
• Avoid sharing passwords or private keys in messages, screenshots or repositories.
• Disconnect from remote sessions when finished.
• Be careful with remote delete, rename and overwrite operations.
• Keep the SSH server and client software updated.

# 17. How this relates to your Windows 11 + WSL Ubuntu

WSL Ubuntu is a Linux environment running on your Windows computer. It is not automatically a separate remote server. Windows can usually browse WSL files using the WSL network share, for example:

```
\\wsl$\Ubuntu\home\your-linux-username
```

That local Windows-to-WSL file access does not require SSH or FileZilla. FileZilla becomes useful when you need to transfer files to a different remote server. To practise SSH locally, you would need an SSH server running in the destination environment and a network route to it.

# 18. A realistic server workflow

1. Obtain the server address, username and approved login method.
2. Connect using SSH and verify the server identity.
3. Use FileZilla with SFTP to upload files to a directory you can write to.
4. Return to SSH to install dependencies or run deployment commands.
5. Check application/service status and logs.
6. Download generated output or logs with SFTP if needed.

# 19. SSH / FTP / SFTP / SCP comparison

| Tool / protocol | Main job | Typical port / transport |
|---|---|---|
| SSH | Remote terminal and command execution | TCP 22 by default |
| FTP | File transfer | TCP 21 control; data connections also used |
| FTPS | FTP protected with TLS | Commonly TCP 21 for explicit mode; configuration varies |
| SFTP | Secure file operations over SSH | TCP 22 by default |
| SCP | Command-line file copy over SSH | SSH transport, usually TCP 22 |

# 20. Quick command reference

| Task | Command |
|---|---|
| Connect to server | ssh user@host |
| Connect to custom SSH port | ssh -p 2222 user@host |
| Verbose connection check | ssh -v user@host |
| Generate key pair | ssh-keygen -t ed25519 |
| Copy public key | ssh-copy-id user@host |
| Upload with SCP | scp file.txt user@host:/remote/path/ |
| Download with SCP | scp user@host:/remote/path/file.txt . |
| Copy folder with SCP | scp -r folder user@host:/remote/path/ |
| Start SFTP | sftp user@host |
| End remote session | exit |

# 21. Practice exercises

**Exercise 1 — Understand a command:** In `ssh ubuntu@192.0.2.10`, identify the command, username and host.

**Exercise 2 — Direction:** If you drag a file from FileZilla's left panel to its right panel, is it an upload or download?

**Exercise 3 — Protocol choice:** You need an encrypted file transfer to a Linux server with SSH enabled. Which FileZilla protocol should you choose?

**Exercise 4 — SCP:** Write a command to upload `project.zip` to `/home/ubuntu/` on a server, using username `ubuntu`.

**Exercise 5 — Troubleshooting:** SSH login succeeds, but FileZilla cannot upload into `/var/www/html`. What should you check? Think about the remote account's directory permissions and ownership; do not immediately use chmod 777.

# 22. Answers for self-check

1. `ssh` is the command; `ubuntu` is the username; `192.0.2.10` is the example host.
2. Upload: local (left) → remote (right).
3. SFTP - SSH File Transfer Protocol.
4. `scp project.zip ubuntu@server_address:/home/ubuntu/`
5. Check whether the remote account has write and execute permission on the destination directory; ask the administrator if elevated access is required.

# 23. Final revision checklist

- ■ Explain local computer vs remote server.
- ■ Define SSH and its usual port 22.
- ■ Distinguish password and key-based authentication.
- ■ Know the difference between private and public keys.
- ■ Explain host-key verification.
- ■ Distinguish FTP, FTPS and SFTP.
- ■ Identify FileZilla's local and remote panels.
- ■ Explain upload and download directions.
- ■ Recognise SSH, SCP and SFTP commands.
- ■ Know why a successful login does not guarantee upload permission.
- ■ Remember that FileZilla is not needed for ordinary local Windows-to-WSL file browsing.

**What is enough for now?** Understand the concepts, practise reading the commands and know how to connect with SFTP when you are given a real server. Advanced SSH server hardening, firewall design, port forwarding and enterprise key management can be studied later.

End of expanded study notes